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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) REGULAREXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCZGY300,

MG5DSCZIM300 -ANIMAL DIVERSITY CHORDATA - II

(2024 ADMISSION ONWARDS)

Duration:1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Part A.MCQ Answer all questions. Each question carries 1marks

1. Which of the following is an endemic bird of the Western Ghats? [K] / [CO3]

a). Malabar Grey Hornbill

2. Which feature supports the high survival rate of Eutherians? [U] / [CO1]

b). Placental nourishment

3. The fusion of the radius and ulna in Frog is associated mainly with: [K] / [CO2]

b). Strengthening of the forelimb

4. Which adaptation helps birds in migratory flight? [K] / [CO3]

a). Fat reserves **(if attempted, give marks)**

5. The three auditory ossicles of mammals are [U] / [CO1]

a). Malleus, incus and stapes

6. Which pairs of ribs are called floating ribs? [U] / [CO2]

d). Last 3 pairs

7. What is the structural status of the syrinx in members of Superorder Palaeognathae?

a). It is completely absent. [K] / [CO1]

8. The young one of kangaroo is commonly called [K] / [CO1]

d). Joey **(if attempted, give marks)**

9. Select the relaxation phase of the heart in rabbit [K] / [CO2]

a). Diastole

10. The roof of the buccal cavity of Rabbit is formed by [K] / [CO2]

b). Palate

11. What is the functional significance of the pecten found inside a bird's eye?

d). Concerned with high power of accommodation. [U] / [CO1]

12. What is the main nitrogenous waste in Rabbit? [An] / [CO2]

a). Urea **[1x12 = 12]**

Part B. Short Answer Questions. Answer any 6 questions
Each question carries 3 marks

13. Compare Delphinus and Dugongs based on adptations [U] / [CO1]

Delphinus (dolphins) and dugongs are aquatic mammals, they have different adaptations based on their distinct diets, habitats and ecological roles

Delphinus **is a Carnivorous predator** feeding on fishes and squid. They use sharp teeth to grab prey and hunt cooperatively. **Streamlined** fusiform body, a **stabilizing dorsal fin** and a **powerful tail fluke** helps to **move fast.** They rely **on echolocation** and interpret echoing sound waves, allowing them to navigate in water. They also have **sharp eyesight**. They have **blowhole on top of the head,** so they can breathe efficiently while swimming at high speeds.

Dugong is a **bottom-dwelling coastal herbivore.** They have **a heavy, blimp-like body** with **no dorsal fin and slow moving.** They **lack echolocation entirely**. They have **sensitive bristles** (*vibrissae*) on their snout to navigate and locate seagrass due to **poor eyesight.** They have **Valved nostrils on top o**f the snout which can be closed tightly when submerged, so they came to the surface to breathe frequently. **(Any2 differences-3 marks)**

14. Describe the optic lobe in pigeons. (3 points-3 marks) [U] / [CO2]

- In Pigeons, two rounded optic lobes are present in midbrain, connected by a transverse optic commissure.
- The two lobes are pushed towards the sides by the enlarged cerebrum and cerebellum.
- Optic lobes are large in size, in correlation with the keen sense of sight, essential for flying and feeding.
- The right and left optic nerves arising from the optic lobes cross over each other and fuse together, forming the optic chiasma.

15. Explain the importance of the cloaca in pigeon. (3 points- 3 marks) [U] / [CO2]

- Cloaca is a common chamber for digestive, urinary and reproductive systems.
- Opens to the outside by the cloacal opening or vent guarded by a muscular sphincter, called anal sphincter.
- In young birds, a glandular sac opens to cloaca posteriorly, called bursa of Fabricius. It is a lymphatic organ, concerned with the maturation of B-cells.
- Cloaca function as a copulatory organ and the transfer of sperms to female is affected by mere cloacal contact.

16. List the major sense organs with present in vertebrates. [K] / [CO2]

Vertebrate receptor organs **(List Any 3-3 marks)**

A. Exteroceptors

1. Photoreceptors - visual organs- Eyes.

2. Stato-acoustic receptors - balancing and hearing organs- Internal ears.

3. Chemoreceptors receptors of chemical stimuli- Nasal mucosa, oral epithelium.

(a) Olfactoreceptors - smelling organs- Nasal mucosa.

(b) Gustatoreceptors - taste receptors- Taste buds of oral epithelium.

4. Cutaneous receptors - receptors in skin- Specialised nerve endings in skin.

(a) Tangoreceptors - touch and pressure receptors.

(b) Thermoreceptors -temperautre receptors.

5. Algesireceptors - pain receptors- Free nerve endings in outer dermis.

B. Interoceptors- Nerve endings which perceive internal stimuli

C. Proprioceptors- Receptors in skeletal muscles, muscle tendons, etc.

17. Describe the main divisions of the nervous system in rabbit [U] / [CO2]

The nervous system of rabbit has three broad divisions.

(1) Central nervous system, including brain and spinal cord;
(2) Peripheral nervous system, including cranial nerves and spinal nerves
(3) Autonomic nervous system consisting sympathetic and parasympathetic nerves.

18. Give a note on the arterial system of Euphlyctis. (brief account only) [U] / [CO2]

The arterial system in frog begins with the truncus arteriosus. The truncus divides into left and right branches. Each branch now divides into three major vessels (1) The common carotid arch supplies to the head region, is a short vessel and divides into two branches, external and internal carotid (2) The systemic arch supplies to body and viscera is the longest of the three and has the greatest distribution of blood. The two systemic arches curve dorsally and join with each other behind the heart to form a common dorsal aorta which gives off many branches supplying to different parts of the body. (3) Pulmocutaneous arch divides into two main branches, pulmonary to the lung and cutaneous to the skin.

19. Explain the function of the oviduct in birds. (3 points- 3 marks) [U] / [CO2]

- **Oviducal funnel or infundibulum** captures the released ovum from the ovary, serves as the site for **fertilization** by stored sperm and adds the first thin layer of albumen.
- Secretion of the thick **albumen** (egg white) around the yolk.
- Secretes the shell membranes.
- Secretes the hard **calcium carbonate** egg shell.
- Secretes the protective outer coating of the shell called the **cuticle** to seal shell pores against bacteria,
- Secretes mucus to aid egg laying and provide muscular contractions to lay the egg.

20. Name the principal bones of the forelimb of a pigeon. [K] / [CO2]

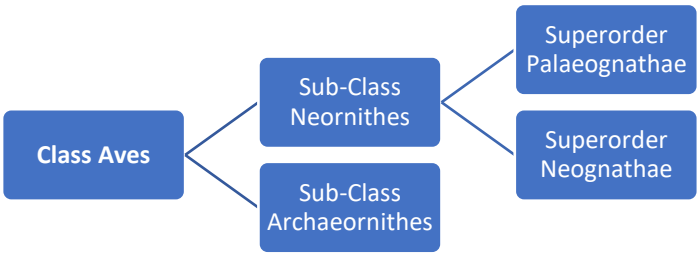
Humerus, Radius, Ulna, Carpals, (proximal row contains two carpals, namely radiale and ulnare) **Metacarpals.** (The three distal carpals in the adult fuse with the three metacarpals, forming a compound bone, called carpo-metacarpus) and **phalanges**. (if main 6 bones are written,0.5 marks each=3 marks) **[3x6 = 18]**

Part C. Short Essay Questions

Answer any 4 questions. Each question carries 5 marks

21.Outline the broad classification of Class Aves into subclasses and super orders and state any three general features of the subclass comprising extinct birds. [K] / [CO1]

(Classification-2 Marks; Any 3 features-3 marks)



Sub-Class Archaeornithes comprises the oldest known fossil birds of the Jurassic period. The salient features are the following:

- Long tail, provided with biserially arranged rectrices and supported by 18-20 caudal vertebrae, without a pygostyle.
- Very primitive wings, with little powers for true flight.
- Free carpals and metacarpals and three clawed fingers.
- Curved and enamelled teeth, set in sockets of both the jaws.

- Amphicoelous vertebrae.
- Keelless sternum.
- Slender thoracic ribs, without uncinate processes.
- Presence of abdominal ribs.
- Example: Archaeopteryx

22.Compare the forelimbs of Frog, Pigeon and Rabbit with respect to their functions. [An]/[CO2]

Animal	Primary Functions	Structural Adaptations -2 points each
Frog	**Body support**	• Supporting the body while resting on land. • Radius and ulna are fused into a single **radio-ulna** bone to provide extra structural strength for weight support.
Pigeon	**Flight**	• Completely modified into **wings** to lift the body and propulsion through the air. Wings are equipped with long flight feathers known as remiges • Humerus, radius and ulna are well developed. Radius and ulna are distinct, parallel to each other and joined only at their ends. The bones are pneumatic (hollow) to reduce body weight. • Distal bones (carpals and metacarpals) are reduced in number to four and fused into a **carpometacarpus** to support flight feathers. The hand has three digits only.
Rabbit	**Burrowing, running and grasping**	• Adapted for quadrupedal locomotion (running) and digging underground burrows for shelter. Equipped with **sharp, strong claws** on the digits to scrape away soil effectively. • Forelimbs retain five distinct, fully mobile digits to assist in grasping food or manipulating objects.

23. Explain the various sensory adaptations of aquatic mammals. [K] / [CO3]

- A few long hairs called bristles are seen at the tip of the snout, sensory in function.
- Eyes are small in size and protected by transparent nictitating membrane.
- They possess a special type of gland called **harderian glands.** The fatty secretion of this gland protects the eye from invasion of foreign germs.
- Dolphins show a special system of communication while they are under water. They can send ultrasonic sound waves which can be received by the other members of the same species which are at long distances.
- They rely on echolocation and interpret echoing sound waves, allowing them to navigate in water. **(5 points-5 marks)**

24.Differentiate the respiratory mechanisms of frog, pigeon and rabbit. [U] / [CO2]

Respiratory mechanisms of pigeon: (3 marks)

Birds make use of two different respiratory mechanisms, one at rest and the other during flight. In any case, inspiration is less active than expiration

In a resting bird, respiratory movements are brought about by ribs and stemum. During inspiration, the intercostal muscles (muscles in between ribs) contract. As a result, the stemum is moved down and the ribs are elevated. This releases the pressure over the lungs and at the same time increases the volume of the body cavity. Consequently, fresh air from outside enters the lungs and reaches the mesobronchus. From there a part of it flows to the inspiratory air sacs. The rest flows to the endobronchi and parabronchi, reaches the bronchioles and gets distributed all over the lungs. Gaseous exchange takes place in the parabronchi and bronchioles.

(if proper diagrammatic representation of double respiration is given only, give marks)
During expiration, sternum and ribs are raised by the contraction of abdominal muscles. As a result, volume of the body cavity decreases and the lungs and inspiratory air sacs get pressed against the ribs. Consequently, used air from the bronchioles is forced out via ectobronchi, primary bronchi and trachea. At the same time, inspiratory air sacs act as bellows to force air from them back to the pulmonary air tubes. Most of the unused air from them flows to the expiratory sacs through recurrent bronchi. A portion of it will flow to the endobronchi and reach the bronchioles. During the next inspiration, unused air from the expiratory sacs will pass to the bronchioles through endobronchi and parabronchi. Parabronchi and bronchioles receive fresh air during inspiration and expiration, and no residual air is retained anywhere in the lungs. So, within the lungs renewal of air and exchange of gases take place twice during every respiratory cycle, one during inspiration and the other during expiration. This kind of respiration is called double respiration. It supplies maximum amount of oxygen, for intense metabolism, complete oxidation and increased energy release.
During flight, air can automatically rush in through the nostrils while the bird flies. The up and down strokes of the wings and the movements of the flight muscles slightly move the vertebral column up and down. This may cause changes in the body cavity and thereby help the ventilation of lungs and air sacs.
Mechanism of respiration in rabbit (2 marks)
Lungs are enclosed in air tight pleural chambers in the thoracic cavity. The movement of ribs and sternum increases the thoracic cavity from side to side and from above downward and the straightening of the diaphragm causes its expansion from before backwards. When the thoracic cavity thus increases in all the three dimensions, pleural cavities and lungs expand, creating a low pressure inside the lungs. This causes the rushing of atmospheric air into the lungs, causing inspiration. During expiration the external intercostal muscles and the muscles of the diaphragm relax and the internal intercostal muscles contract. As a result, ribs, sternum and diaphragm return to their original position. This decreases the volume of the thoracic and pleural cavities and the lungs, causing expiration.
(If respiratory mechanisms of pigeon and rabbit is written, give marks)

25.Explain the digestive system of pigeon and compare it with that of frog. [U] / [CO2]

Digestive system comprises the alimentary canal and the glands associated with it. Alimentary canal is a longcoiled tube and differentiated into several functionally different regions, such as buccal cavity, pharynx, oesophagus, crop, stomach, duodenum, ileum and rectum which opens to the cloaca. Buccal cavity is bounded above and below by the toothless horny beak. On the buccal floor is the pointed tongue. It contains mucous glands and scanty taste buds. Buccal roof is formed by the palate, which is a partition between the buccal and nasal cavities. Salivary glands are absent. But there are a few mucus secreting buccal glands. Their secretion moistens the food. Buccal cavity is followed by the pharynx. Pharynx leads to the long, tubular and distensible oesophagus or gullet. It passes down the neck and terminally enlarges to a thin-walled sac or reservoir, known as crop. The crop serves three purposes, namely (i) storage of large amount of ingested food to enable the bird to feed quickly (ii) maceration (softening) of the food by storage (iii) moistening of the food. Pigeons are unique in producing a white proteinaceous and fatty secretion, called 'pigeon's milk' or 'crop milk'. It is produced in both the sexes by a pair of crop glands during breeding season and used to nourish the squabs until they are able to feed upon grains. Crop leads to the stomach. Stomach has two portions, fore-stomach or Proventriculus is the tubular and glandular initial part concerned with enzyme production and gizzard or hind-stomach is the thick-walled and muscular terminal part, serves as a muscular mill for crushing and grinding the food. It is equipped with a horny internal lining with tooth-like ridges for crushing the food. Gizzard leads to the small intestine through a small

passage called pylorus, guarded by a pyloric sphincter. Small intestine has two parts, duodenum is a short and bent part and ileum is a long and coiled tube. In between the two duodenal limbs lies the pancreas. Joining the duodenum is a bile ducts and pancreatic ducts. The internal lining of the ileum is produced into numerous villi to increase the absorptive area. Following the ileum is the short rectum that represents the large intestine. At the junction of ileum and rectum is a pair of blind pouches, called rectal caeca or colic caeca. Terminally, rectum opens to the cloaca. This opening is guarded by a muscular sphincter, called anal sphincter. Cloaca has three linear compartments, namely coprodaeum, urodaeum and proctodaeum (or vestibule). Coprodaeum receives the rectum, urodaeum receives the urinary and genital ducts, and proctodaeum opens to the outside by the cloacal opening or vent.

Digestive glands: Buccal glands, gastric glands, intestinal glands, liver and pancreas are the major glands associated with the alimentary canal. Buccal glands are not true salivary glands, secrete enzyme-free mucus to moisten the food. Gastric glands are the glandular pits in the mucous membrane of proventriculus. They secrete gastric juice, which contains HCl and the proteolytic enzyme pepsin. Intestinal glands are seen in the mucous membrane of ileum. They secrete intestinal juice, which contains proteolytic, amylolytic and lipolytic enzymes. Liver is a bilobed gland, secretes the alkaline bile. Gall bladder is absent in adult pigeons. Pancreas is a heterocrine gland, with exocrine secretion is called pancreatic juice. It is discharged to the the duodenum through three pancreatic ducts. Pancreatic juice contains protein-splitting, carbohydrate-splitting and fat-splitting enzymes.

Digestive system of Frog: Frog has a very wide mouth bordered by immovable lips. Protrusible, glandular sticky tongue is used in capturing the prey. In frogs, the base of the tongue is attached anteriorly at the margin of the jaw and the free end is folded back on the floor of the mouth when at rest. Teeth are confined to the upper jaw and vomers. Lower jaw is toothless. Dentition is polyphyodont, homodont and acrodont. The oesophagus is extremely short. In frog the cardiac end of the stomach is wide and the pyloric end is short and narrow. Small intestine is coiled and large intestine is short and straight. Villi appear for the first time in amphibians. An ileocolic valve is present between small and large intestines.The large intestine opens into the cloaca.

Digestive glands: Only mucous glands are present in the buccal cavity. Salivary glands are absent. Liver is large and well developed. Gall bladder is present. Pancreas is compact and lies in the mesentery stretching between the duodenum and the stomach.

(Digestive system of pigeon-3marks; frog -2 mark)

26.Explain how the skin of a Frog helps in cutaneous respiration and water balance. [U] / [CO2] (Any 5 points-5 marks)

- Skin is smooth, slimy and glandular with mucus glands
- Skin is naked without exoskeleton/scales.
- Skin is richly supplied with blood capillaries that lies directly beneath the surface and is permeable to gases.
- Before oxygen can diffuse into blood it must first be dissolved in water and that is why frogs always stay near water to keep their skin moist. It is further kept moist by secretion of mucus from its glands.
- cutaneous respiration operates continuously serving as the primary method all the time when a frog is underwater or hibernating.
- Frogs absorb water directly through their skin. The skin on a frog's lower belly and thighs—often called the pelvic patch—is specially adapted with high permeability to rapidly pull in moisture from damp soil, bodies of water.
- Mucus secretions helps to slow down excessive evaporation, if frogs kept in dry air for too long. **[5x4 = 20]**